

Taylor-Informed Predictive Cost Adaptive Control for Quadrotors with Online Gravity-Trim Adaptation

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Abstract—This paper develops Taylor-informed predictive cost adaptive control (PCAC) for quadrotors with online gravity-trim adaptation. First-, second-, and third-order expansions of the nonlinear dynamics about nominal hover define sparse sampled-data dictionaries for row-wise recursive least-squares identification with variable-rate forgetting. At each step, the identified predictor is linearized at the current state, and its Jacobian is fixed over the prediction horizon. The identified vertical dynamics also estimate the vehicle mass and gravity-trim input, eliminating fixed nominal gravity compensation. Simulations with an abrupt payload change and aggressive helix tracking show that the higher-order predictors improve prediction and tracking while preserving the standard PCAC formulation.

I. INTRODUCTION

Quadrotors are a representative nonlinear control problem because they combine underactuation, strong coupling, and real-time constraints [1]–[3]. Model uncertainty, payload changes, and disturbances motivate adaptive flight-control methods that maintain performance without repeated offline identification [4]–[7].

Model predictive control (MPC) provides a systematic treatment of tracking objectives and constraints [8]. Adaptive MPC combines predictive optimization with online identification [9]–[11]. Predictive cost adaptive control (PCAC), in particular, identifies a sampled-data predictor and constructs the local prediction model directly from the identified dynamics [12]–[14]. Related work has extended adaptive prediction to nonlinear and behavioral models [15], [16].

Payload changes alter the vehicle mass and corresponding hover thrust. A fixed nominal gravity-compensation term therefore produces steady-state error when the true mass differs from its nominal value. Although adaptive flight control can estimate changing parameters [17]–[19], existing quadrotor PCAC relies on a sparse hover-linear predictor [20], whose prediction accuracy degrades during larger excursions from hover.

This paper develops a nonlinear formulation based on local Taylor structure. First-, second-, and third-order expansions about nominal hover generate sparse sampled-data dictionaries for row-wise recursive least-squares identification. The identified vertical dynamics provide online mass and gravity-trim estimates. At each control step, the nonlinear predictor is linearized at the current state and held fixed over the horizon, yielding an affine model for the standard PCAC optimization.

Simulations evaluate an abrupt payload change and aggressive helix tracking. Higher-order predictors improve tracking and one-step prediction, and the trim estimate removes the need for fixed nominal compensation. The complete MATLAB implementation

is available at <https://github.com/tamwng/taylor-informed-pcac-quadrotor>.

II. PROBLEM SETUP

Let Σ_i and Σ_b denote the inertial and body-fixed frames, respectively, and let $e_j \in \mathbb{R}^3$, $j \in \{1, 2, 3\}$, be the Euclidean basis vectors. The inertial e_3 -axis points upward. The position and velocity in Σ_i are $p \in \mathbb{R}^3$ and $v := \dot{p}$. The attitude $R := {}^iR_b \in SO(3)$ maps body-frame components to inertial coordinates, and $\omega \in \mathbb{R}^3$ is the body-frame angular velocity. Let $f \geq 0$ be the total thrust along the body e_3 -axis and $\tau \in \mathbb{R}^3$ the body torque. The dynamics are

$$\dot{p} = v, \quad (1)$$

$$m\dot{v} = -mge_3 + fRe_3, \quad (2)$$

$$\dot{R} = R\omega^\times, \quad (3)$$

$$J\dot{\omega} = -\omega^\times J\omega + \tau, \quad (4)$$

where $m > 0$, $J = J^\top \succ 0$, and $g > 0$ denote the mass, inertia, and gravitational acceleration. For a body-frame vector a_b , $a_i = Ra_b$.

For $\omega = [\omega_1 \ \omega_2 \ \omega_3]^\top$, the skew-symmetric map $(\cdot)^\times : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ is defined by

$$\omega^\times := \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}. \quad (5)$$

Parameterize the attitude by the intrinsic Z - X - Y angles $\xi := [\psi \ \varphi \ \theta]^\top$, where ψ , φ , and θ are yaw, roll, and pitch. Then $R(\xi) = R_z(\psi)R_x(\varphi)R_y(\theta)$. Using $c_\alpha := \cos \alpha$ and $s_\alpha := \sin \alpha$ yields

$$R(\xi) = \begin{bmatrix} c_\psi c_\theta - s_\psi s_\theta & -s_\psi c_\theta & c_\psi s_\theta + s_\psi c_\theta \\ s_\psi c_\theta + s_\psi s_\theta & c_\psi c_\theta & s_\psi s_\theta - s_\psi c_\theta \\ -c_\psi s_\theta & s_\psi & c_\psi c_\theta \end{bmatrix}. \quad (6)$$

The angular velocity satisfies $\omega = E(\xi)\dot{\xi}$, where

$$E(\xi) = \begin{bmatrix} -s_\theta c_\varphi & c_\theta & 0 \\ s_\varphi & 0 & 1 \\ c_\theta c_\varphi & s_\theta & 0 \end{bmatrix}. \quad (7)$$

Hence, for $c_\varphi \neq 0$, $\dot{\xi} = E^{-1}(\xi)\omega$, where

$$E^{-1}(\xi) = \begin{bmatrix} -\frac{s_\theta}{c_\varphi} & 0 & \frac{c_\theta}{c_\varphi} \\ c_\theta & 0 & s_\theta \\ s_\theta \tan \varphi & 1 & -c_\theta \tan \varphi \end{bmatrix}. \quad (8)$$

Define the state and input as

$$x := [p^\top \ \xi^\top \ v^\top \ \omega^\top]^\top \in \mathbb{R}^{12}, \quad (9)$$

$$u := [f \ \tau^\top]^\top \in \mathbb{R}^4. \quad (10)$$

Equations (1)–(4) become

$$\dot{x} = \mathcal{F}(x, u; m, J), \quad (11)$$

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where

$$\mathcal{F}(x, u; m, J) = \begin{bmatrix} v \\ E^{-1}(\xi)\omega \\ -ge_3 + \frac{f}{m}R(\xi)e_3 \\ J^{-1}(-\omega^\times J\omega + \tau) \end{bmatrix}. \quad (12)$$

III. NOMINAL HOVER COORDINATES AND SAMPLED DYNAMICS

For any $p_e \in \mathbb{R}^3$, the hover equilibrium for mass m is

$$x_e = [p_e^\top \quad 0_{1 \times 3} \quad 0_{1 \times 3} \quad 0_{1 \times 3}]^\top, \quad (13)$$

$$u_e(m) = [mg \quad 0 \quad 0 \quad 0]^\top. \quad (14)$$

For nominal mass m_0 , set $u_{e,0} := [m_0g \ 0 \ 0 \ 0]^\top$. The pair $(x_e, u_{e,0})$ is not an equilibrium when $m \neq m_0$.

Define the shifted variables $\tilde{x} := x - x_e$, $\tilde{u} := u - u_{e,0}$. The shifted dynamics are

$$\begin{aligned} \dot{\tilde{x}} &= \bar{\mathcal{F}}(\tilde{x}, \tilde{u}; m, m_0, J) \\ &:= \mathcal{F}(x_e + \tilde{x}, u_{e,0} + \tilde{u}; m, J). \end{aligned} \quad (15)$$

At the nominal hover point,

$$\bar{\mathcal{F}}(0, 0; m, m_0, J) = \begin{bmatrix} 0_{3 \times 1} \\ 0_{3 \times 1} \\ g(\frac{m_0}{m} - 1)e_3 \\ 0_{3 \times 1} \end{bmatrix}. \quad (16)$$

The mass mismatch therefore appears as a constant vertical acceleration.

Under zero-order hold with sampling period $T_s > 0$, $u(t) = u_k$, $t \in [kT_s, (k+1)T_s)$, denote the flow map of $\mathcal{F}(\cdot, \cdot; m, J)$ by $F_{m,J}$:

$$x_{k+1} = F_{m,J}(x_k, u_k). \quad (17)$$

With full-state feedback, the shifted sampled-data dynamics are

$$\tilde{x}_{k+1} = \bar{F}_{m,m_0,J}(\tilde{x}_k, \tilde{u}_k), \quad (18)$$

where

$$\bar{F}_{m,m_0,J}(\tilde{x}, \tilde{u}) := F_{m,J}(x_e + \tilde{x}, u_{e,0} + \tilde{u}) - x_e. \quad (19)$$

Define $\beta_g(m, m_0) := g(\frac{m_0}{m} - 1)$. At $(\tilde{x}, \tilde{u}) = (0, 0)$, the sampled-data residual is

$$\bar{F}_{m,m_0,J}(0, 0) = \begin{bmatrix} \frac{T_s^2}{2}\beta_g(m, m_0)e_3 \\ 0_{3 \times 1} \\ T_s\beta_g(m, m_0)e_3 \\ 0_{3 \times 1} \end{bmatrix}. \quad (20)$$

Consequently,

$$\bar{F}_{m,m_0,J}(0, 0) = 0 \iff m = m_0. \quad (21)$$

IV. TAYLOR EXPANSION ABOUT THE NOMINAL HOVER POINT

Partition the shifted input as

$$\tilde{u} = [\tilde{f} \quad \tilde{\tau}^\top]^\top, \quad \tilde{\tau} = [\tilde{\tau}_1 \quad \tilde{\tau}_2 \quad \tilde{\tau}_3]^\top. \quad (22)$$

Because the nominal torque is zero, $\tilde{\tau} = \tau$, while $f = m_0g + \tilde{f}$.

Define $\alpha_0(m, m_0) := \frac{m_0g}{m}$, $\alpha_f(m) := \frac{1}{m}$, and let $z := [\tilde{x}^\top \ \tilde{u}^\top]^\top$. Expand about $z = 0$, corresponding to $(x_e, u_{e,0})$. The Euler-angle representation is local to a neighborhood

with $c_\varphi \neq 0$.

For $D \in \{1, 2, 3\}$, let $\bar{\mathcal{F}}^{[D]}$ be the degree- D Taylor polynomial of $\bar{\mathcal{F}}$ at $z = 0$:

$$\bar{\mathcal{F}}(\tilde{x}, \tilde{u}; m, m_0, J) = \bar{\mathcal{F}}^{[D]}(\tilde{x}, \tilde{u}; m, m_0, J) + r_D(z), \quad (23)$$

where $r_D(z)$ contains terms of total degree greater than D . Write the retained field as

$$\bar{\mathcal{F}}^{[D]}(\tilde{x}, \tilde{u}; m, m_0, J) = \begin{bmatrix} v \\ \eta_D(\xi, \omega) \\ a_D(\xi, \tilde{f}; m, m_0) \\ \Omega_D(\omega, \tilde{\tau}; J) \end{bmatrix}. \quad (24)$$

The position row is exact; the other rows depend on D .

The Euler-kinematic terms are

$$\eta_1(\xi, \omega) = \begin{bmatrix} \omega_3 \\ \omega_1 \\ \omega_2 \end{bmatrix}, \quad (25)$$

$$\eta_2(\xi, \omega) = \eta_1(\xi, \omega) + \begin{bmatrix} -\theta\omega_1 \\ \theta\omega_3 \\ -\varphi\omega_3 \end{bmatrix}, \quad (26)$$

$$\eta_3(\xi, \omega) = \eta_2(\xi, \omega) + \begin{bmatrix} \frac{1}{2}(\varphi^2 - \theta^2)\omega_3 \\ -\frac{1}{2}\theta^2\omega_1 \\ \theta\varphi\omega_1 \end{bmatrix}. \quad (27)$$

The translational acceleration terms are

$$a_1(\xi, \tilde{f}; m, m_0) = \begin{bmatrix} \alpha_0\theta \\ -\alpha_0\varphi \\ \beta_g + \alpha_f\tilde{f} \end{bmatrix}, \quad (28)$$

$$a_2(\xi, \tilde{f}; m, m_0) = a_1(\xi, \tilde{f}; m, m_0) + \begin{bmatrix} \alpha_0\varphi\psi + \alpha_f\tilde{f}\theta \\ \alpha_0\psi\theta - \alpha_f\tilde{f}\varphi \\ -\frac{\alpha_0}{2}(\varphi^2 + \theta^2) \end{bmatrix}, \quad (29)$$

$$a_3(\xi, \tilde{f}; m, m_0) = a_2(\xi, \tilde{f}; m, m_0) + \begin{bmatrix} -\frac{\alpha_0}{6}\theta^3 - \frac{\alpha_0}{2}\psi^2\theta + \alpha_f\tilde{f}\varphi\psi \\ \frac{\alpha_0}{6}\varphi^3 + \frac{\alpha_0}{2}\varphi\psi^2 + \frac{\alpha_0}{2}\varphi\theta^2 + \alpha_f\tilde{f}\psi\theta \\ -\frac{\alpha_f}{2}\tilde{f}(\varphi^2 + \theta^2) \end{bmatrix}. \quad (30)$$

Arguments of α_0 , α_f , and β_g are suppressed in (28)–(30).

The angular-acceleration terms are

$$\Omega_1(\omega, \tilde{\tau}; J) = J^{-1}\tilde{\tau}, \quad (31)$$

$$\Omega_2(\omega, \tilde{\tau}; J) = \Omega_3(\omega, \tilde{\tau}; J) = J^{-1}(-\omega^\times J\omega + \tilde{\tau}). \quad (32)$$

For principal-axis body coordinates, $J = \text{diag}(J_1, J_2, J_3)$ and

$$\Omega_2(\omega, \tilde{\tau}; J) = \Omega_3(\omega, \tilde{\tau}; J) = \begin{bmatrix} \frac{J_2 - J_3}{J_1}\omega_2\omega_3 + \frac{1}{J_1}\tilde{\tau}_1 \\ \frac{J_3 - J_1}{J_2}\omega_3\omega_1 + \frac{1}{J_2}\tilde{\tau}_2 \\ \frac{J_1 - J_2}{J_3}\omega_1\omega_2 + \frac{1}{J_3}\tilde{\tau}_3 \end{bmatrix}. \quad (33)$$

V. SPARSE TAYLOR-INFORMED SAMPLED-DATA DICTIONARY

For $J = \text{diag}(J_1, J_2, J_3)$, the Taylor terms in (25)–(32) define a row-wise sampled-data dictionary. Table I lists the complete degree-one regressors; Table II lists the additional degree-two and degree-three monomials.

For each row, $\phi_{q,D}$ concatenates the table entries through degree D , removes duplicates while preserving their first occurrence, and omits empty entries. Translational-acceleration monomials appear in both the position and velocity rows. For example, the vertical-velocity row uses

$$\phi_{v3,1}(z) = [v_3 \quad 1 \quad \tilde{f}]^\top, \quad (34)$$

$$\phi_{v3,2}(z) = [v_3 \quad 1 \quad \tilde{f} \quad \varphi^2 \quad \theta^2]^\top, \quad (35)$$

$$\phi_{v3,3}(z) = [v_3 \quad 1 \quad \tilde{f} \quad \varphi^2 \quad \theta^2 \quad \tilde{f}\varphi^2 \quad \tilde{f}\theta^2]^\top. \quad (36)$$

TABLE I
COMPLETE DEGREE-ONE ROW REGRESSORS.

State row q	$\phi_{q,1}(z)$
$\tilde{p}_1$	$[\tilde{p}_1 \ v_1 \ \theta \ \omega_2 \ \tilde{\tau}_2]^\top$
$\tilde{p}_2$	$[\tilde{p}_2 \ v_2 \ \varphi \ \omega_1 \ \tilde{\tau}_1]^\top$
$\tilde{p}_3$	$[\tilde{p}_3 \ v_3 \ 1 \ \tilde{f}]^\top$
ψ	$[\psi \ \omega_3 \ \tilde{\tau}_3]^\top$
φ	$[\varphi \ \omega_1 \ \tilde{\tau}_1]^\top$
θ	$[\theta \ \omega_2 \ \tilde{\tau}_2]^\top$
v_1	$[v_1 \ \theta \ \omega_2 \ \tilde{\tau}_2]^\top$
v_2	$[v_2 \ \varphi \ \omega_1 \ \tilde{\tau}_1]^\top$
v_3	$[v_3 \ 1 \ \tilde{f}]^\top$
ω_1	$[\omega_1 \ \tilde{\tau}_1]^\top$
ω_2	$[\omega_2 \ \tilde{\tau}_2]^\top$
ω_3	$[\omega_3 \ \tilde{\tau}_3]^\top$

TABLE II
ADDITIONAL TAYLOR MONOMIALS RETAINED AT DEGREES TWO AND THREE.

State row q	Degree-two addition $\delta\phi_{q,2}(z)$	Degree-three addition $\delta\phi_{q,3}(z)$
$\tilde{p}_1, v_1$	$[\varphi\psi, \tilde{f}\theta]^\top$	$[\theta^3, \psi^2\theta, \tilde{f}\varphi\psi]^\top$
$\tilde{p}_2, v_2$	$[\psi\theta, \tilde{f}\varphi]^\top$	$[\varphi^3, \varphi\psi^2, \varphi\theta^2, \tilde{f}\psi\theta]^\top$
$\tilde{p}_3, v_3$	$[\varphi^2, \theta^2]^\top$	$[\tilde{f}\varphi^2, \tilde{f}\theta^2]^\top$
ψ	$[\theta\omega_1]^\top$	$[\varphi^2\omega_3, \theta^2\omega_3]^\top$
φ	$[\theta\omega_3]^\top$	$[\theta^2\omega_1]^\top$
θ	$[\varphi\omega_3]^\top$	$[\varphi\theta\omega_1]^\top$
ω_1	$[\omega_2\omega_3]^\top$	$\emptyset$
ω_2	$[\omega_3\omega_1]^\top$	$\emptyset$
ω_3	$[\omega_1\omega_2]^\top$	$\emptyset$

Order the state rows as

$$\mathcal{Q} := (q_1, \dots, q_{12}) = (\tilde{p}_1, \tilde{p}_2, \tilde{p}_3, \psi, \varphi, \theta, v_1, v_2, v_3, \omega_1, \omega_2, \omega_3). \quad (37)$$

For each $q \in \mathcal{Q}$ and $D \in \{1, 2, 3\}$, define $n_{q,D} := \dim \phi_{q,D}$. The identified row model is

$$\hat{F}_{q,D,k}(z) = \hat{\Theta}_{q,D,k}^\top \phi_{q,D}(z), \quad \hat{\Theta}_{q,D,k} \in \mathbb{R}^{n_{q,D}}. \quad (38)$$

Stacking the row models gives

$$\hat{F}_D(z) := \text{col}_{i=1}^{12} \left(\hat{F}_{q_i,D,k}(z) \right). \quad (39)$$

The three dictionaries contain

$$\sum_{q \in \mathcal{Q}} n_{q,1} = 40, \quad (40)$$

$$\sum_{q \in \mathcal{Q}} n_{q,2} = 58, \quad (41)$$

$$\sum_{q \in \mathcal{Q}} n_{q,3} = 80. \quad (42)$$

Degree one is the sparse hover-linear model with a vertical trim intercept. Degree two adds thrust-attitude, Euler-kinematic, and gyroscopic couplings; degree three adds cubic thrust-direction and Euler-kinematic terms.

VI. TAYLOR-INFORMED RECURSIVE IDENTIFICATION

Apply RLS independently to each state row. For fixed $D \in \{1, 2, 3\}$ and $q \in \mathcal{Q}$, define

$$\varphi_{q,D,k} := \phi_{q,D}(z_k) \in \mathbb{R}^{n_{q,D}}. \quad (43)$$

The one-step prediction

$$\hat{\tilde{x}}_{q,k+1|k} = \hat{\Theta}_{q,D,k}^\top \varphi_{q,D,k}, \quad (44)$$

has error

$$e_{q,D,k+1} := \tilde{x}_{q,k+1} - \hat{\tilde{x}}_{q,k+1|k}. \quad (45)$$

Initialize

$$\hat{\Theta}_{q,D,0} \in \mathbb{R}^{n_{q,D}}, \quad (46)$$

$$P_{q,D,0} = p_{q,0} I_{n_{q,D}}, \quad p_{q,0} > 0. \quad (47)$$

For each transition $(z_k, \tilde{x}_{k+1})$, update

$$L_{q,D,k} = \lambda_k^{-1} P_{q,D,k}, \quad (48)$$

$$P_{q,D,k+1} = L_{q,D,k} - L_{q,D,k} \varphi_{q,D,k} \varphi_{q,D,k}^\top L_{q,D,k} \times (1 + \varphi_{q,D,k}^\top L_{q,D,k} \varphi_{q,D,k})^{-1} \varphi_{q,D,k}^\top L_{q,D,k}, \quad (49)$$

$$\hat{\Theta}_{q,D,k+1} = \hat{\Theta}_{q,D,k} + P_{q,D,k+1} \varphi_{q,D,k} e_{q,D,k+1}, \quad (50)$$

The forgetting factor $\lambda_k \in (0, 1]$ is computed from the stacked error

$$e_{D,k} := \text{col}_{q \in \mathcal{Q}} (e_{q,D,k}) \quad (51)$$

using the variable-rate forgetting (VRF) rule in [21]; $\lambda_k = 1$ disables forgetting.

Let $\hat{c}_{D,k}$ and $\hat{b}_{D,k}$ be the coefficients of the constant and $\tilde{f}$ in the vertical-velocity row. They yield

$$\hat{\beta}_{g,k} = \frac{\hat{c}_{D,k}}{T_s}, \quad (52)$$

$$\hat{\alpha}_{f,k} = \frac{\hat{b}_{D,k}}{T_s}, \quad (53)$$

$$\hat{m}_k = \frac{1}{\hat{\alpha}_{f,k}}, \quad (54)$$

$$\hat{f}_{\text{trim},k} = -\frac{\hat{\beta}_{g,k}}{\hat{\alpha}_{f,k}}, \quad (55)$$

$$\hat{f}_{\text{trim},k} = m_0 g + \hat{f}_{\text{trim},k}. \quad (56)$$

Project $\hat{\alpha}_{f,k}$ onto a positive admissible interval before computing $\hat{m}_k$ and $\hat{f}_{\text{trim},k}$.

VII. PREDICTIVE COST ADAPTIVE CONTROL

For fixed $D \in \{1, 2, 3\}$, linearize the identified map at the current operating point:

$$z_k := [\tilde{x}_k^\top \ \tilde{u}_k^\top]^\top, \quad (57)$$

$$\left. \frac{\partial \hat{F}_{D,k}}{\partial z} \right|_{z=z_k} = [A_{D,k} \ B_{D,k}], \quad (58)$$

$$c_{D,k} := \hat{F}_{D,k}(z_k) - A_{D,k} \tilde{x}_k - B_{D,k} \tilde{u}_k. \quad (59)$$

For $i = 0, \dots, N-1$, hold this Jacobian fixed over the horizon:

$$\tilde{x}_{i+1|k} = c_{D,k} + A_{D,k} \tilde{x}_{i|k} + B_{D,k} \tilde{u}_{i|k}. \quad (60)$$

Define the estimated trim inputs

$$\hat{u}_{\text{trim},k} := [\hat{f}_{\text{trim},k} \ 0 \ 0 \ 0]^\top, \quad (61)$$

$$\hat{\tilde{u}}_{\text{trim},k} := \hat{u}_{\text{trim},k} - u_{e,0}. \quad (62)$$

Parameterize the predicted input by its deviation from trim:

$$\tilde{u}_{i|k} = \hat{u}_{\text{trim},k} + \delta u_{i|k}. \quad (63)$$

Substitution into (60) gives

$$\tilde{x}_{i+1|k} = d_{D,k} + A_{D,k}\tilde{x}_{i|k} + B_{D,k}\delta u_{i|k}, \quad (64)$$

where $d_{D,k} := c_{D,k} + B_{D,k}\hat{u}_{\text{trim},k}$.

Following the PCAC timing convention, initialize

$$\tilde{x}_{0|k} = \tilde{x}_k, \quad (65)$$

$$\delta u_{0|k} = u_k - \hat{u}_{\text{trim},k}, \quad (66)$$

and define

$$\Delta \delta u_{i|k} := \delta u_{i|k} - \delta u_{i-1|k}, \quad i = 1, \dots, N-1. \quad (67)$$

Lift (64) over the horizon:

$$X_{1|k} = \Gamma_{D,k}\tilde{x}_k + T_{D,k}U_{1|k} + C_{D,k}, \quad (68)$$

where

$$X_{1|k} := \text{col}(\tilde{x}_{1|k}, \dots, \tilde{x}_{N|k}), \quad (69)$$

$$U_{1|k} := \text{col}(\delta u_{1|k}, \dots, \delta u_{N-1|k}), \quad (70)$$

$$\Delta U_{1|k} := \text{col}(\Delta \delta u_{1|k}, \dots, \Delta \delta u_{N-1|k}), \quad (71)$$

$$E_{1|k} := \text{col}(\varepsilon_{1|k}, \dots, \varepsilon_{N|k}), \quad (72)$$

$$\hat{U}_{\text{trim},k} := \text{col}(\hat{u}_{\text{trim},k}, \dots, \hat{u}_{\text{trim},k}). \quad (73)$$

The lifted matrices $\Gamma_{D,k}$, $T_{D,k}$, and $C_{D,k}$ follow [12].

For shifted reference $\tilde{r}_{i|k}$, solve

$$\begin{aligned} \min_{U_{1|k}, E_{1|k}} \quad & \frac{1}{2} \sum_{i=1}^N \|\tilde{x}_{i|k} - \tilde{r}_{i|k}\|_{Q_i}^2 \\ & + \frac{1}{2} \sum_{i=1}^{N-1} \|\Delta \delta u_{i|k}\|_R^2 + \frac{1}{2} \sum_{i=1}^N \|\varepsilon_{i|k}\|_S^2 \end{aligned} \quad (74)$$

$$\begin{aligned} \text{s. t.} \quad & X_{1|k} = \Gamma_{D,k}\tilde{x}_k + T_{D,k}U_{1|k} + C_{D,k}, \\ & \mathcal{H}_x X_{1|k} \leq h_x + E_{1|k}, \\ & \mathcal{U}_{\min} \leq \hat{U}_{\text{trim},k} + U_{1|k} \leq \mathcal{U}_{\max}, \\ & \Delta \mathcal{U}_{\min} \leq \Delta U_{1|k} \leq \Delta \mathcal{U}_{\max}, \\ & E_{1|k} \geq 0, \end{aligned}$$

where $Q_i \succeq 0$, $R \succ 0$, and $S \succ 0$; the constraint notation follows [12].

Apply the first optimized deviation at step $k+1$:

$$u_{k+1} = \hat{u}_{\text{trim},k} + \delta u_{1|k}^*. \quad (75)$$

VIII. NUMERICAL EVALUATION

A. Simulation Setup

The exact nonlinear plant (11)–(12) is integrated by fixed-step RK4 with 10 substeps per sampling interval and zero-order-held input. Twelve scalar RLS estimators, initialized from the exact zero-order-hold discretization of the nominal hover-linear model, identify the sampled-data map and share the VRF factor [21]. At each sample, the identified map is linearized and the PCAC problem is solved by the quadprog active-set algorithm with at most 180 iterations and constraint tolerance 10^{-7} .

The public MATLAB implementation reproduces the reported cases through `main_pcac_quadrotor.m`; all simulation and controller parameters are centralized in

TABLE III
SIMULATION PARAMETERS.

Parameter	Value
T_s	0.10 s
N	10
D	$\{1, 2, 3\}$
m_0	4.34 kg
g	9.81 m/s ²
J	diag(0.0820, 0.0845, 0.1377) kg m ²
$m_0 g$	42.5754 N
$u_{\min}$	$[0, -2, -2, -1.2]^\top$
$u_{\max}$	$[85, 2, 2, 1.2]^\top$
$\Delta u_{\max}$	$[6, 0.35, 0.35, 0.25]^\top$
$\Delta u_{\min}$	$-\Delta u_{\max}$
Attitude bounds	$ \psi \leq \pi, \phi , \theta \leq 50^\circ$
Q	diag(110, 110, 150, 2, 3, 3, 18, 18, 28, 0.7, 0.7, 0.7)
Q_N	$8Q$
R	diag(0.015, 0.18, 0.18, 0.12)
S	$10^5 I_6$
$P_{q,D,0}$	0.1I
(τ_n, τ_d)	(5, 25)
η	0.50
Mass projection	[2.5, 7.0] kg

`pcac_parameters.m`, the reference trajectories are defined in `generate_reference.m`, and the reported metrics are computed in `performance_metrics.m`.

Case 1 imposes an abrupt mass change to test the mass and trim estimates. Case 2 tracks an aggressive three-dimensional helix at constant mass to test the higher-order predictors away from hover. Both cases use $x(0) = 0_{12}$, $u(0) = [m_0 g, 0, 0, 0]^\top$, noise-free full-state measurements, and zero applied force and torque disturbances. All Taylor degrees use the same plant, tuning, constraints, and horizon. Table III lists the parameters.

The first component of each input-bound vector is thrust in newtons; the remaining components are torques in newton-meters. Both cases use the common 3-s reference ramp

$$s(t) = \begin{cases} 0, & t \leq 0, \\ \frac{1}{2} [1 - \cos(\frac{\pi t}{3})], & 0 < t < 3, \\ 1, & t \geq 3. \end{cases} \quad (76)$$

In both cases, the velocity reference is the analytical derivative of the position reference, and the attitude and angular-rate references are zero. For K samples, Tables IV–V use

$$\text{RMSE}_p = \sqrt{\frac{1}{K} \sum_{k=1}^K \|p_k - r_{p,k}\|_2^2}, \quad (77)$$

$$\text{RMSE}_{\text{pred}} = \sqrt{\frac{1}{K-1} \sum_{k=2}^K \|e_{p,k}\|_2^2}, \quad (78)$$

$$\text{Input variation} = \sum_{k=1}^{K-1} \|u_{k+1} - u_k\|_1, \quad (79)$$

$$\text{Maximum tracking error} = \max_k \|p_k - r_{p,k}\|_2, \quad (80)$$

where $e_{p,k}$ denotes the position components of the one-step prediction error.

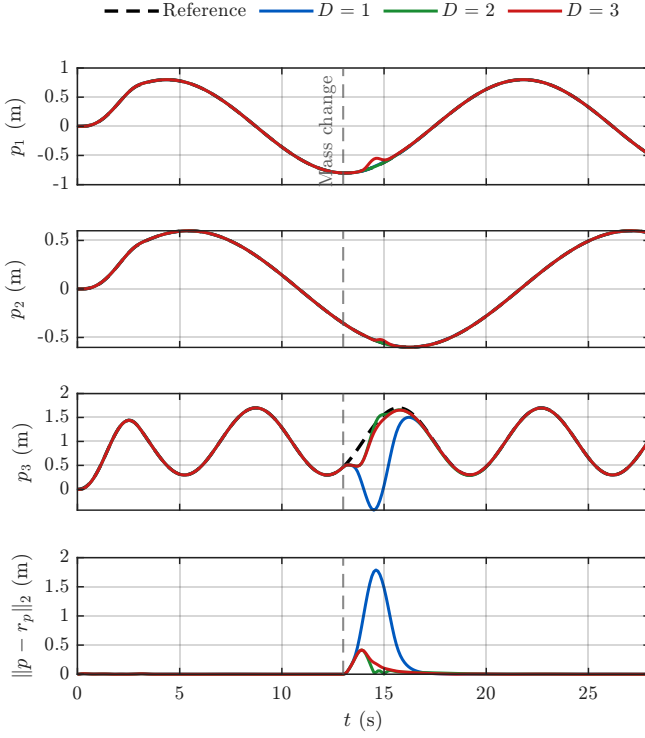


Fig. 1. **Case 1.** Position tracking after the mass change at $t = 13$ s; the lower panel shows $\|p - r\|_2$.

TABLE IV

CASE 1: ABRUPT MASS-CHANGE PERFORMANCE (4.34 $\rightarrow$ 5.60 KG).

D	Pos. RMSE (m)	Final $ m - \hat{m} $ (kg)	Final $ f_{\text{trim}} - \hat{f}_{\text{trim}} $ (N)
1	0.348	0.052	0.028
2	0.062	0.066	0.017
3	0.069	0.031	0.007

B. Case 1: Abrupt Mass Change

The case ends at $t = 28$ s. Its position reference is

$$r_{p,1} = 0.80s(t) \sin(0.36t), \quad (81)$$

$$r_{p,2} = 0.60s(t) \sin(0.29t), \quad (82)$$

$$r_{p,3} = s(t) [1 + 0.70 \sin(0.90t)]. \quad (83)$$

The mass changes from $m_0 = 4.34$ kg to $m = 5.60$ kg at $t = 13$ s. Figs. 1 and 2 show the tracking response, online estimates, and VRF factor; Table IV reports the errors.

Both higher-order predictors reduce the tracking error. Degree two gives the lowest position RMSE, whereas degree three gives the most accurate terminal mass and trim estimates.

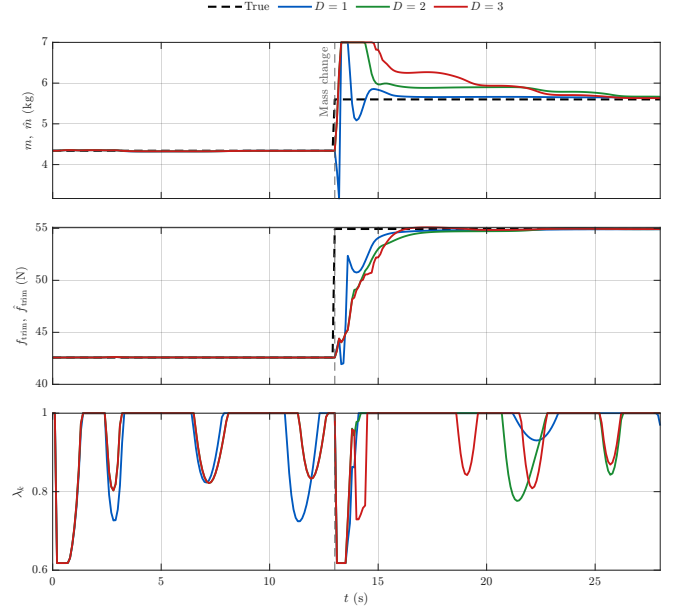


Fig. 2. **Case 1.** Mass and trim-thrust estimates and VRF factor after the 4.34–5.60 kg mass change. Dashed curves denote true values.

TABLE V

CASE 2: AGGRESSIVE HELIX TRACKING.

D	Pos. RMSE (m)	Pred. RMSE (m)	Input var.	Max. track. err. (m)
1	0.0519	0.001281	154.73	0.1695
2	0.0288	0.000559	69.27	0.1664
3	0.0281	0.000483	59.41	0.1664

C. Case 2: Aggressive Helix Tracking

The quadrotor tracks a three-dimensional helix at constant mass $m = m_0 = 4.34$ kg until $t = 21$ s, with

$$\chi(t) = 1.50(t - 3), \quad (84)$$

$$r_{p,1} = 1.50s(t)[\cos \chi(t) - 1], \quad (85)$$

$$r_{p,2} = 1.50s(t) \sin \chi(t), \quad (86)$$

$$r_{p,3} = 0.80s(t) + 0.075 \int_0^t s(\sigma) d\sigma + 0.22s(t) \sin(0.78t). \quad (87)$$

Figure 3 shows the trajectories, while Fig. 4 shows the one-step prediction error and attitude. Table V reports the performance metrics.

Degrees two and three provide similar tracking accuracy, but degree three gives the lowest one-step prediction error and control variation.

— — Reference — $D = 1$ — $D = 2$ — $D = 3$

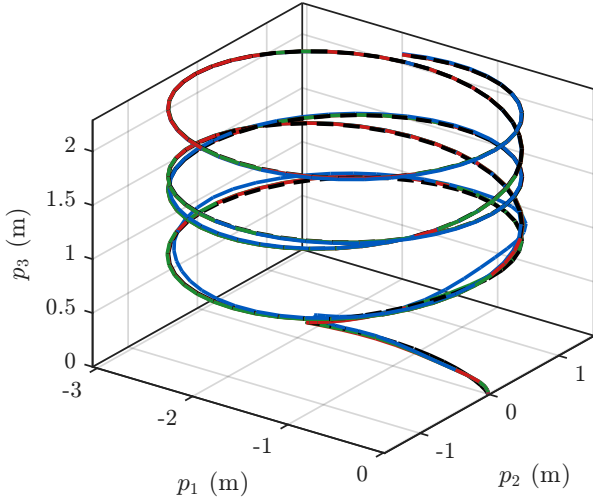


Fig. 3. **Case 2.** Helical reference and closed-loop trajectories.

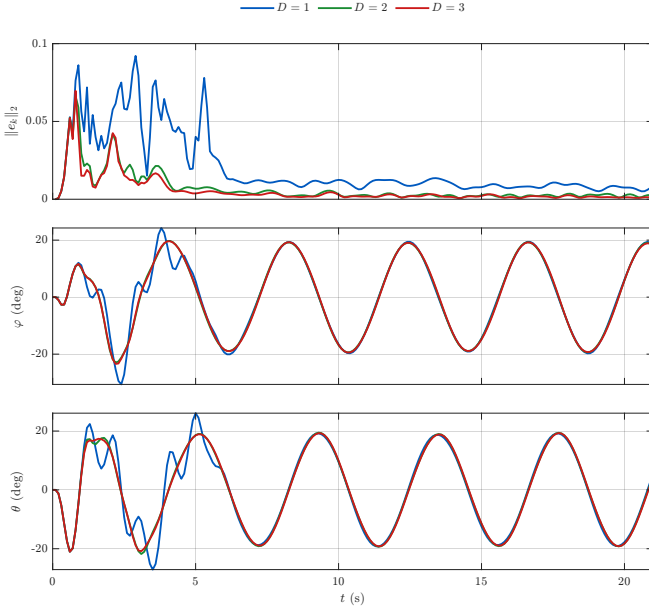


Fig. 4. **Case 2.** One-step prediction-error norm and attitude responses during helix tracking.

IX. CONCLUSION

Taylor-informed PCAC combines sparse higher-order predictors with online identification and gravity-trim adaptation. The second- and third-order models improved tracking and one-step prediction relative to the first-order model, while the identified vertical dynamics recovered the mass and trim input after an abrupt payload change. These results show that local higher-order structure can improve PCAC away from hover without changing its optimization framework.

Future work will address motion-capture experiments and globally nonsingular attitude representations.

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REFERENCES

- [1] T. Lee, M. Leok, and N. H. McClamroch, “Geometric tracking control of a quadrotor UAV on $SE(3)$,” in *49th IEEE Conference on Decision and Control (CDC)*. IEEE, 2010, pp. 5420–5425.
- [2] H. Rastgoftar and I. V. Kolmanovsky, “Safe affine transformation-based guidance of a large-scale multi-quadcopter system,” *IEEE Transactions on Control of Network Systems*, vol. 8, no. 2, pp. 640–653, 2021.
- [3] E. Hermard, T. W. Nguyen, M. Hosseinzadeh, and E. Garone, “Constrained control of UAVs in geofencing applications,” in *2018 26th Mediterranean Conference on Control and Automation (MED)*. IEEE, 2018, pp. 217–222.
- [4] Z. T. Dydek, A. M. Annaswamy, and E. Lavretsky, “Adaptive control of quadrotor UAVs: A design trade study with flight evaluations,” *IEEE Transactions on Control Systems Technology*, vol. 21, no. 4, pp. 1400–1406, 2013.
- [5] R. J. Richards, J. A. Marshall, and D. S. Bernstein, “Experimental flight testing a quadcopter autopilot based on predictive cost adaptive control,” in *2025 American Control Conference (ACC)*. IEEE, 2025, pp. 2471–2476.
- [6] T. He, S. Burton, C. Spencer, and W. Wei, “Projection-based online parameter estimation of a tilt-rotor VTOL aircraft and experimental validation,” *ASME Letters in Dynamic Systems and Control*, vol. 6, no. 2, p. 021012, 2026.
- [7] J. C. Vander Schaaf, K. J. Fidkowski, and D. S. Bernstein, “Active flow control using adaptive model predictive control with online, closed-loop system identification,” *Journal of Guidance, Control, and Dynamics*, vol. 49, no. 7, pp. 1883–1896, 2026.
- [8] U. Eren, A. Prach, B. B. Koçer, S. V. Raković, E. Kayacan, and B. Açikmeşe, “Model predictive control in aerospace systems: Current state and opportunities,” *Journal of Guidance, Control, and Dynamics*, vol. 40, no. 7, pp. 1541–1566, 2017.
- [9] D. W. Clarke, C. Mohtadi, and P. S. Tuffs, “Generalized predictive control—part I: The basic algorithm,” *Automatica*, vol. 23, no. 2, pp. 137–148, 1987.
- [10] —, “Generalized predictive control—part II: Extensions and interpretations,” *Automatica*, vol. 23, no. 2, pp. 149–160, 1987.
- [11] R. Tao, P. Zhao, I. V. Kolmanovsky, and N. Hovakimyan, “Robust adaptive MPC in the presence of nonlinear time-varying uncertainties: An uncertainty compensation approach,” *Automatica*, vol. 192, p. 113142, 2026.
- [12] T. W. Nguyen, S. A. U. Islam, D. S. Bernstein, and I. V. Kolmanovsky, “Predictive cost adaptive control: A numerical investigation of persistency, consistency, and exigency,” *IEEE Control Systems Magazine*, vol. 41, no. 6, pp. 64–96, 2021.
- [13] R. J. Richards, S. A. U. Islam, and D. S. Bernstein, “Predictive cost adaptive control of the NASA benchmark flutter model,” *Journal of Guidance, Control, and Dynamics*, vol. 48, no. 12, pp. 2663–2679, 2025.
- [14] A. Farahmandi and B. Reitz, “Predictive cost adaptive control of a planar missile with unmodeled aerodynamics,” in *AIAA SCITECH 2024 Forum*. AIAA, 2024.
- [15] T. W. Nguyen, “Adaptive behavioral predictive control: State-free regulation without hankel weights,” 2026. [Online]. Available: <https://arxiv.org/abs/2602.12016>
- [16] R. A. Alhazmi, A. S. Babu, S. A. U. Islam, and D. S. Bernstein, “Nonlinear predictive cost adaptive control of pseudo-linear input-output models using polynomial, Fourier, and cubic spline observables,” 2026. [Online]. Available: <https://arxiv.org/abs/2602.05263>
- [17] B. Lai and D. S. Bernstein, “Adaptive Kalman filtering developed from recursive least squares forgetting algorithms,” in *2024 American Control Conference (ACC)*. IEEE, 2024, pp. 4378–4383.
- [18] A. Burtsev, A. Jariwala, E. Bakolas, and D. Goldstein, “Adaptive feedback flow control for wings,” *AIAA Journal*, pp. 1–14, 2026.
- [19] Z. Huang, M. Chen, and H. Shen, “FLS-based adaptive flight control of fixed-wing UAV with augmented switching model,” *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 56, no. 3, pp. 2036–2048, 2026.
- [20] T. W. Nguyen, “Fast RLS identification leveraging the linearized system sparsity: Predictive cost adaptive control for quadrotors,” in *2026 American Control Conference (ACC)*. IEEE, 2026.
- [21] A. L. Bruce, A. Goel, and D. S. Bernstein, “Convergence and consistency of recursive least squares with variable-rate forgetting,” *Automatica*, vol. 119, p. 109052, 2020.